National University of Computer and Emerging Sciences
Lahore Campus

Object Oriented Data Structures
(CS2021)
Date: 29-October-2026
Course Instructor(s):
Dr. Fatima Hameed Khan
Miss Khadija Naveed

Sessional-II
Total Time: 1 Hour(s)
Total Marks: 30
Total Questions: 02

Student Name _____ Roll No _____ Section _____ Signature _____

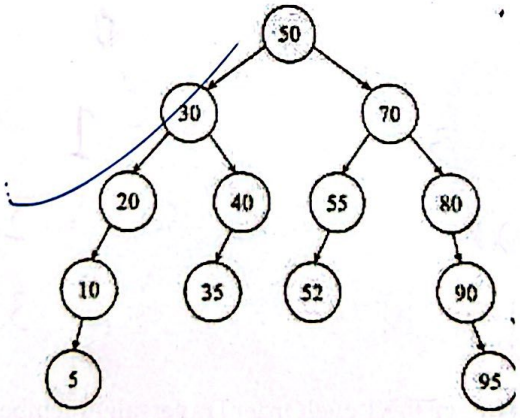
Please answer all parts of a question together on the answer sheet.

CLO #3 (PLO 3, C6): **Design tree and graph data structures and their associated algorithms.**

Question 1 [Marks 5+3+4]

a. **Construct** the **void BST::Insert (DT value)** member function in C++ for a Binary Search Tree (BST) that inserts a new node at its appropriate position while maintaining the Binary Search Tree property. Your implementation should handle insertion into both an empty and a non-empty tree. The BST class is the same as you have studied in your theory and lab classes.

b. Given the following Binary Search Tree



Construct the resulting Binary Search Tree after deleting the node with data value 30 showing all the steps for full credit.

c. **Generate** the output of preorder traversal when carried out on the binary tree in part (b).

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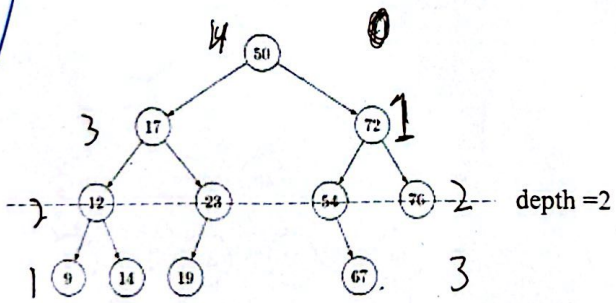
Question 2 [Marks 10+8]

Suppose that we have a Binary Tree class defined as

```
class BNode{
public: // the below data members are made public for ease during exam.
        int key;
        BNode *left, *right;
};

class BinaryTree{
private:
        BNode *root;
public:
/*Standard Binary Tree member functions*/
    int NoOfNodesWithDepth(int d); //function to implement recursively
    void LevelOrderTraversal();//function to implement iteratively
};
```

i. You are required to **program** in C++ the NoOfNodesWithDepth(int d) member function recursively (using only the ADT provided above, no other data structure is allowed) that counts the number of nodes which have the depth greater than input parameter "d". The depth of the root node is 0.
So, if this function is called for the following binary tree with value of d passed as 2, it will return 4 as the total number of nodes having depth greater than 2 is 4.



ii. You are required to **construct** the LevelOrderTraversal() member function iteratively in C++ (with no worse than O(n) performance) that print all nodes present in the binary tree.

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